Integrated Environmental Assessment and Management, 2025, **21(4)**, 735–738
https://doi.org/10.1093/inteam/vjaf059
Advance access publication: May 21, 2025
Letter to the Editor

Research gaps and recommendations to improve the Safe and Sustainable by Design framework

Annegaaike Leopold[1], Michelle Bloor[2,*], Ksenia Groh[3], Leo Posthuma[4], Hans Sanderson[5], Hanna Schreiber[6], Christoph Schür[7], and Paul Thomas[8]

[1]Calidris Environment BV, Warnsveld & Dutch Board for the Authorisation of Plant Protection Products and Biocides (Ctgb), The Netherlands
[2]University of Glasgow, Glasgow, United Kingdom
[3]Eawag—Swiss Federal Institute of Aquatic Science and Technology, Switzerland
[4]RIVM, The Netherlands & Department of Environmental Science, Radboud University, The Netherlands
[5]Aarhus University, Denmark
[6]Environment Agency Austria, Austria
[7]Eawag—Swiss Federal Institute of Aquatic Science and Technology, Switzerland
[8]KREATiS, France

*Corresponding author: Michelle Bloor. Email: Michelle.Bloor@glasgow.ac.uk

Introduction

The "Safe and Sustainable by Design" (SSbD) vision, highlighted in the European Commission's (EC) Chemicals Strategy for Sustainability (CSS), part of the European Green Deal, is central to ensuring the long-term sustainability of the chemical sector and its products in Europe (Caldeira et al., 2022; EC, 2020). The framework marks a paradigm shift in the regulatory management of chemical manufacturing and use, in that it promotes a proactive assessment of both safety and sustainability throughout the innovation process, including its early stages. To enable effective implementation of the SSbD framework, the key concepts and approaches used to perform safety and sustainability assessments need to be aligned and improved further, both scientifically and practically. The multifaceted nature of SSbD calls for joint co-development and testing of novel approaches and methodologies, where a multipartite involvement is key: for example, what can be offered by science (academia and beyond) needs to match what can be operationalized in practice (industry, business), and SSbD overall needs to facilitate the ultimate societal goals such as the toxic-free environment (government).

The Society of Environmental Toxicology and Chemistry (SETAC) hosts a multisectoral membership possessing highly relevant expertise for the discussion of forward-looking aspects of SSbD principles and practices. The SETAC Europe sounding board, tasked with supporting SETAC's contributions to the EC's High Level Roundtable on the Implementation of the CSS, held a series of SSbD-focused consultations in 2023–2024. These were organized in response to a request from, and in collaboration with, the EC's Directorate-General for Research and Innovation. These consultations aimed to identify opportunities and research gaps to improve the scientific basis and the operational utility of the SSbD framework, focusing mainly on the environmental and partly on the social aspects, but not covering the governance aspects. In this letter, we communicate the major outcomes of the stakeholder consultations, seeking to support the SSbD initiation, testing, and optimization processes, thus complementing the EC's SSbD guidance (Abbate et al., 2024). Although the SSbD framework originated in the European context, the findings are of broader relevance for the global practices of chemical assessment and management.

Consultation process

The organization of stakeholder consultations followed a stepwise approach. In the first step, an extensive list of potential ideas was compiled. These were sorted and prioritized in the second step and used to inform the discussions during the final step, where the main conceptual opportunities and research gaps were identified and reflected on. Further details of the consultation process and the original expert contributions received in each step are compiled in three respective reports, available through SETAC resource links (see online supplementary material S1, S2, and S3; see also Leopold et al., 2024).

Conceptual opportunities

Two main conceptual opportunities were recognized. First, a forward-looking vision towards repurposing the results and insights obtained in the context of SSbD research can prove beneficial beyond the design of chemicals per se. Policy developments in Europe have led to a series of regulations that focus on advancing the zero pollution goal and ensuring the long-term sustainability of the chemical sector in Europe (e.g., the Corporate Sustainability Due Diligence Directive [CSDDD], the Corporate Sustainability Reporting Directive [CSRD], and the Sustainable Finance Disclosure Regulation [SFDR]). Although these regulations may have distinct implementation pathways, the SSbD research themes presented below can also be operationalized more broadly, in multiple instruments and contexts. Although the current Omnibus Package I (https://commission.europa.eu/news/commission-proposes-cut-red-tape-and-simplify-business-environment-2025-02-26_en) does affect the CSRD and SFDR, neither

Received: April 16, 2025. **Revised:** May 01, 2025. **Accepted:** May 20, 2025
© The Author(s) 2025. Published by Oxford University Press on behalf of the Society of Environmental Toxicology and Chemistry.
This is an Open Access article distributed under the terms of the Creative Commons Attribution License (https://creativecommons.org/licenses/by/4.0/), which permits unrestricted reuse, distribution, and reproduction in any medium, provided the original work is properly cited.

Downloaded from https://academic.oup.com/ieam/article/21/4/735/8139937 by Lib4RI - Library of Eawag, Empa, PSI, WSL user on 11 July 2025

the CSS nor SSbD are directly affected by this—the Competitive Compass of Europe maintains a focus on developing a new chemical industry package (https://commission.europa.eu/priorities-2024-2029/competitiveness_en). Hence, there continues to be a need to support companies as they optimize their environmental performance and comply with the various policy objectives and regulations, thereby providing the European chemical industry with a unique selling point for the global market, which enhances their competitiveness. For policy-making, this would embody an extra advantage for the "one substance, one assessment" approach. Chemicals designed under the SSbD framework would thus not only meet the SSbD requirements, but the data generated would also be valuable for other initiatives, such as the Product and Organisation Environmental Footprinting (PEF/OEF) relevant for chemical companies and the financial sector.

The second conceptual opportunity concerns the prioritization of SSbD efforts towards the most promising areas of sustainability impact and improvement. Focusing on things that matter most instead of equally considering all 16 impact categories of the current environmental footprint approach (EC, 2021) would be more practical. Indeed, impact categories relevant for different products may differ vastly; for example, although land and water use, carbon dioxide emissions, and the pesticide pollution footprint will likely matter most for a cotton T-shirt, some other impact categories, such as those stemming from mining, will likely be identifiable as dominant when considering electric car batteries. Further examples of such decisive impact categories include climate change and resource use. Methods must be developed to find the main impact categories that drive the sustainability score or classification of substances or products. Examples can be found in pollution research, where impacts per site are always dominated by a few chemicals. Similarly, researchers have identified product-specific sustainability metrics that can be used as proxies for predicting damage to human health and biodiversity (Steinmann et al., 2017). These observations can be systematically substantiated further to help prioritize cases and/or applications where SSbD-guided assessment and innovation could prove to be most effective in an initial screening and prioritization phase.

Research gaps

The stakeholder consultations identified seven research gaps that can be divided into three categories: (i) data generation needs (research gaps 1 and 2), (ii) methodological frameworks needs (research gaps 3 and 4), and (iii) co-creation needs (research gaps 5, 6, and 7).

Research gap 1

The first research gap relates to generation of empirical data and the development of high-throughput new approach methodologies (NAMs) to fill current voids. The SSbD assessments need more data than are currently available, and closing data gaps for early-stage innovation is key. New approach methodologies can provide latitude for this. The assessment and comparison of the safety and sustainability of chemicals and materials in SSbD should include a proper consideration of both acute and long-term chronic effects. However, although the acute toxicity NAMs are often already available, the development of fit-for-purpose chronic toxicity NAMs capable of replacing in vivo animal endpoints (e.g., growth inhibition or reprotoxicity) requires further research and wider recognition. Additionally, there is a lack of NAMs for accurately predicting degradation in a time and cost-efficient way that is needed to rapidly inform innovation processes, especially where mixtures are concerned. There is also a need for NAMs that can address upscaling extrapolation to ecosystems and landscapes in an adverse outcome pathway context, providing a mechanistic explanation for observed effects.

Robust NAMs, both in silico and in vitro, should be considered. These can be used, independently or together, at different technology readiness levels (TRLs) and thus serve to inform the innovation process. A faster evaluation/qualification/validation process should be proposed to ensure the assessments are based on state-of-the-art science and can thus help to reduce the gap between science and regulatory application. This can be further supported by ensuring adherence to FAIR (findable, accessible, interoperable, and reusable) data management principles. In conjunction with the opportunities and characteristics of well-designed tiered approaches (see research gap 3), research should seek to enhance consistency in the results of SSbD assessments, typically undertaken across several innovation steps, up to the point of the final decision-making. The TRLs that apply to this research gap are considered to be 2–5. The TRLs are broad indications of distance to realization. The higher the TRL level, the less time is estimated to be needed until implementation. The lowest is TRL 1, and TRL 9 is the highest. The TRL numbers shown in the text aim to indicate the approximate position of discussed ideas on the TRL ladder, as estimated by the experts involved in the consultation.

Research gap 2

The second research gap relates to generation of computational data, including the development and validation of AI for SSbD purposes. The use of AI approaches to fill data gaps, particularly within the early-stage innovation assessments, is a clear opportunity, relevant for both the safety and sustainability impact metrics. Initial results obtained for hazard assessment show that this specific form of NAMs, capable of using the body of existing (environmental/eco) toxicity test data to provide insights into missing data on SSbD-relevant aspects, may hold enormous potential for advancing data generation for SSbD assessment purposes. It is, however, critical that the AI-based tools in SSbD research yield reliable outcomes, which require the consideration of data quality and model validation. Additionally, further development of novel AI-based principles, approaches, and tools is needed to support access to generation and integration of data for safety and sustainability impact metrics as well as ensuring adherence to FAIR principles. The TRLs that apply here are considered to be 2–4.

Research gap 3

The third research gap relates to advancing the next-generation PEF/OEF methodology (EC, 2021) and its adaptation to the chemical sector, including the adoption of tiered assessment frameworks. As discussed above in relation to the second conceptual opportunity, robust approaches to prioritize PEF impact categories for novel chemicals or products are urgently needed. More refined weighting of key drivers of impact would enable better differentiated sustainability scoring, whereby accuracy and validity need to be confirmed by case studies and suitable evaluation processes. To handle data-poor situations typical of early stages of innovation, a tiered data requirement and analysis system (including but not limited to NAMs-based testing) should be developed for each impact category. The requirements for each tier should be proportional, with indicators ranging from fast and conservative (relevant for small and medium enterprises and early innovation stages) to comprehensive and accurate (for large companies and later "ready to market" stages). Because

Downloaded from https://academic.oup.com/ieam/article/21/4/735/8139937 by Lib4RI - Library of Eawag, Empa, PSI, WSL user on 11 July 2025

chemicals often co-occur as mixtures in products and processes, and ultimately, in the end-of-life phase and in nature, this research should also address mixtures. The TRLs are considered to range between 2–5.

Research gap 4

The fourth research gap relates to the development of next-generation life cycle assessment (LCA) methodologies. The SSbD assessment should cover all relevant LCA components in a balanced and effective way, which can be attained by integrating the tiered approaches classically used in risk assessment (whereby the TRL categories would relate to the increasingly refined tiers). Furthermore, to improve relevance and comprehensiveness of SSbD assessments, novel LCAs should be developed or optimized for key areas of interest. In addition to the social LCA, these should address, for example: greenhouse gas emissions, waste and pollution, biodiversity, water, and circularity. The LCAs should range from cradle-to-gate level (resource extraction to factory gate) all the way to cradle-to-cradle level, which is required in the innovation process and the product. Life cycle inventory and costing tools should also be adapted as needed. Along with further development of LCA methodologies, tools, and databases, ultimately, a systems approach is required in which the LCA-guided contributions to SSbD assessments are connected instead of operating in isolation. The TRL range for this research gap is considered to be 5–8.

Research gap 5

The fifth research gap concerns the need for widespread co-creation efforts to test and adapt SSbD assessment methods and outcomes to increase efficacy in supporting the decision-making related to the sustainability transition processes in the chemical sector. An effective, comprehensive, and validated approach requires further adoption and broad implementation of the optimized SSbD assessment tools, which can be supported by research using practical case studies. Co-creation partners and stakeholders involved in the case studies should include large and small chemical companies and manufacturers along with environmental, social, and governance accounting and audit experts, other representatives of the financial sector, and SSbD experts from academia, regulatory authorities, and nongovernmental organizations. This research gap thus identifies the need for not only continuing but also intensifying cross-sectoral connections and co-creation initiatives. The associated TRLs are considered to be 5–7.

Research gap 6

The sixth research gap arises from insufficient understanding of the socioeconomic factors that influence reciprocal interactions between chemicals and people. In the context of equity, diversity, and inclusion considerations, multi- and transdisciplinary research is needed to understand the divergent impacts of chemicals on different communities and how different people would address and solve health and environmental problems with chemicals, depending on (a) ancestries, (b) gender identities, (c) life stories, and (d) socioeconomic driven differential exposure. Based on the co-created research insights into socioecological systems, the societal interpretation of SSbD requirements and results may differ and could result in the establishment of unusual "solution pathways" and divergent timelines, depending, e.g., on the regional aspects that can influence both exposure to threats and access to training. Therefore, instead of an assumed "one-size-fits-all" use of SSbD principles and outcomes, it should be acknowledged that different solution pathways may yield optimal results under different conditions. The TRLs for this research gap are considered to be 3–5.

Research gap 7

The seventh and final research gap deals with co-creation in connection with education, training, and skills. There is a clear need for the development of international masters and PhD programs that integrate horizontal knowledge and understanding across disciplines related to chemical safety assessment, including hazard, exposure, and risk assessment; LCA; and SSbD. Cross-sectoral learning, such as the SSbD-wide use of tiering to improve the utility of LCA as suggested in research gap 3, is considered highly beneficial for SSbD implementation. In addition, multidisciplinary teams must be developed with experts from sustainability and safety domains originating from industry, government, and academia to co-create the educational outreach on SSbD implementation for professionals and students. This would ensure an efficient build-up of resources, including capability and capacity to translate the SSbD vision into reality. This gap could be the most challenging to address, as it would require changes in established patterns as well as altered interpretations and innovations regarding regulations. It has therefore been allocated a TRL of 7–9.

Conclusion and follow-up

Numerous discussions during the stakeholder consultations have emphasized that a proactive approach to characterizing SSbD assessments is of utmost importance. These must be developed, optimized, validated, and used to safeguard human health and protect and restore environmental quality. To date, the approaches used in policy and practice have not yet led us to a toxic-free environment status, as human health and biodiversity impacts are still ongoing. The broad implementation of the SSbD framework is thus urgent, but also challenging, as it requires the optimal safety and sustainability balance for a wide array of chemicals and situations to be found under data-scarce conditions. In addition to the reflections presented in this article, the authors welcome further discussion and elaboration of the identified opportunities, recommendations, and research gaps. We also look forward to continuing the discussion with the EC and focal points in the individual member states of the EU, to accelerate research to address the identified gaps. This will enhance the implementation of a sound, robust, and accurate SSbD framework that would effectively support a sustainable transition of the chemical sector in Europe under the Green Deal and the EU Competitive Compass. SETAC will, in the meantime, be progressing the science around the SSbD concepts and practice. For example, SSbD topics were discussed during the 35th SETAC Europe Annual Meeting in 2025 and ongoing discussions are being facilitated in SETAC Europe by the authors.

Supplementary material

Supplementary material is available online at *Integrated Environmental Assessment and Management*.

Data availability

The reports of each of the three consultations held are available through open access resource links. These are: i) https://www.setac.org/resource/preliminary-report-of-the-1st-ssbd-consultation-meeting-pdf.html; ii) https://www.setac.org/resource/preliminary-rep-2nd-consultation-meeting_final-pdf.html; iii) https://

Downloaded from https://academic.oup.com/ieam/article/21/4/735/8139937 by Lib4RI - Library of Eawag, Empa, PSI, WSL user on 11 July 2025

www.setac.org/resource/workshop-report-seville_preliminary-insights-from-setac-ec-online-consultations-pdf.html.

Author contributions

Annegaaike Leopold (Conceptualization, Investigation, Methodology, Project administration, Visualization, Writing—original draft, Writing—review & editing), Michelle Bloor (Conceptualization, Investigation, Methodology, Visualization, Writing—original draft, Writing—review & editing), Ksenia Groh (Conceptualization, Investigation, Methodology, Visualization, Writing—original draft, Writing—review & editing), Leo Posthuma (Conceptualization, Investigation, Methodology, Visualization, Writing—original draft, Writing—review & editing), Hans Sanderson (Conceptualization, Investigation, Methodology, Visualization, Writing—original draft, Writing—review & editing), Hanna Schreiber (Conceptualization, Investigation, Methodology, Visualization, Writing—original draft, Writing—review & editing), Christoph Schür (Conceptualization, Investigation, Methodology, Visualization, Writing—original draft, Writing—review & editing), and Paul Thomas (Conceptualization, Investigation, Methodology, Visualization, Writing—original draft, Writing—review & editing)

Funding

The research was undertaken and the manuscript was written without a source of funding.

Conflicts of interest

The authors are all members of the Society of Environmental Toxicology and Chemistry (SETAC). The authors declare no conflict of interest.

Disclaimer

This article presents the outcomes of the SETAC—EC SSbD-consultation series held between 2023 and 2024. It does not represent the positions of SETAC or EC or any of the organizations where the authors are employed.

Acknowledgments

The authors would like to thank the European Commission (EC) colleagues from Directorate-General Research and Innovation and the Joint Research Centre, as well as Bruno Campos, Unilever, for the discussions about the SSbD framework and for their support in setting up the SETAC—EC SSbD-consultation series that led to this publication. In addition, the authors would like to thank the SETAC members who took part in the consultation series and without whom it would not have been possible to develop these recommendations. Finally, we would like to thank the SETAC Office for their logistical support of the consultations.

References

Abbate, E., Garmendia Aguirre, I., Bracalente, G., Mancini, L., Tosches, D., Rasmussen, K., Bennett, M. J., Rauscher, H., & Sala, S. (2024). *Safe and Sustainable by Design chemicals and materials—Methodological guidance*. Publications Office of the European Union. https://doi.org/10.2760/28450, JRC138035. https://publications.jrc.ec.europa.eu/repository/handle/JRC138035.

Caldeira, C., Farcal, R., Garmendia Aguirre, I., Mancini, L., Tosches, D., Amelio, A., Rasmussen, K., Rauscher, H., Riego Sintes, J., & Sala, S. (2022). *Safe and Sustainable by Design chemicals and materials—Framework for the definition of criteria and evaluation procedure for chemicals and material* (EUR 311100 EN). Publications Office of the European Union. http://doi.org/10.2760/487955, JRC128591. https://publications.jrc.ec.europa.eu/repository/handle/JRC128591.

EC. (2020). Communication from the Commission to the European Parliament, the Council, The European Economic and Social Committee and the Committee of the Regions. Chemicals Strategy for Sustainability Towards a Toxic-Free Environment, COM (2020) 667. https://eur-lex.europa.eu/legal-content/EN/TXT/? uri=celex : 52020DC0667.

EC (2021). Commission recommendation of 16.12.2021 on the use of the Environmental Footprint methods to measure and communicate the life cycle environmental performance of products and organisations, COM (2021) 9332. https://environment.ec.europa.eu/document/download/cb899bd7-bb06-491d-9989-c856a401fcd0_en.

Leopold, A., Bloor, M., Groh, K., Posthuma, L., Sanderson, H., Schreiber, H., Schür, C., Thomas, P. (Eds.). (2024). Recommendations and research gaps to improve the function of the Safe and Sustainable by Design (SSbD) concept: SETAC—EC Consultation series 2023-24. https://doi.org/10.5281/zenodo.14228553, preprint: not peer reviewed.

Steinmann, Z. J. N., Schipper, A. M., Hauck, M., Giljum, S., Wernet, G., & Huijbregts, M. A. J.; Resource. (2017). Footprints are good proxies of environmental damage. *Environmental Science & Technology*, 51, 6360–6366. https://doi.org/10.1021/acs.est.7b00698.

© The Author(s) 2025. Published by Oxford University Press on behalf of the Society of Environmental Toxicology and Chemistry.
This is an Open Access article distributed under the terms of the Creative Commons Attribution License (https://creativecommons.org/licenses/by/4.0/), which permits unrestricted reuse, distribution, and reproduction in any medium, provided the original work is properly cited.
Integrated Environmental Assessment and Management, 2025, 21, 735–738
https://doi.org/10.1093/inteam/vjaf059
Letter to the Editor

Downloaded from https://academic.oup.com/ieam/article/21/4/735/8139937 by Lib4RI - Library of Eawag, Empa, PSI, WSL user on 11 July 2025